PRANJAL SWARUP BHATNAGAR

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EXPERIENCE

Associate, Norman Lab Apr 2026 - Present
Next Tech Lab (QS Reimagine Education Award winner) *SRMIST, Kattankulathur*

- Selected by interview into Norman Lab at Next Tech Lab, the first student-led organisation from India to win the QS Reimagine Education Award.
- Built StupidHacks, the portal for the lab's internal hackathon, with a small team. Owned the data layer and its rule tests, the design system on the lab's brand, and the move to Supabase with one-time-code roster sign-in. Live at stupidhack.nexttechlab.in.

Member, campus tech communities Oct 2026 - Present
HackTheBox Chennai (Cybersecurity & Dev), GDG on Campus SRM (Technical), Aaruush (Team Envision) *Chennai*

PROJECTS

ExamDB | **Python, FastAPI, PostgreSQL, Next.js, Docker**

- Built and shipped a searchable database of **194** Indian competitive exams, consolidating dates scattered across dozens of government portals into one source that cites the official page behind every date.
- Designed a scrape/normalise/dedup/reduce pipeline that preserves conflicting sources and flags the disagreement instead of silently picking a winner, so wrong data stays traceable to the scraper that produced it.
- Caught a scraper that passed its own tests while emitting **46** rows of a historical archive stamped with the current year, spotted from an impossible 29 FEB entry. Added source-diff validation so a row count is never mistaken for correctness.
- Built OAuth sign-in storing only SHA-256 digests of opaque session tokens, plus daily scrapers on GitHub Actions and a Next.js frontend. Solo project, live at examdb.org.

TYPE | **Rust, ratatui, LSP, tree-sitter**

- Building a full IDE that runs in the terminal, non-modal with full mouse parity, aimed at the gap between bare terminal editors and heavyweight GUI ones. Published on crates.io as typ-editor (**v0.4**), with static musl builds and installers that verify checksums.
- Wrote LSP and tree-sitter clients so language support arrives by protocol instead of per-language code. Diagnostics, go-to-definition, completion and signature help, with grammars compiled into the binary and parsed off the render thread.
- Holds a **100ms** cold-start and **16ms** p99 keystroke-to-glyph budget, enforced by perf tests in CI. The parallel project walk indexes **37,586** files in **95ms** against 2.6s serial.
- Every action is reachable by key, palette and mouse, and a test fails the build when one lacks a path. An opt-in vim layer maps modes onto the same named actions, so the core never depends on it.

cowbird | **Python, asyncio, curl_cffi, FastAPI, Docker**

- Library, CLI and HTTP server putting **32** disposable-email providers behind one async interface (generate an address, wait for mail, pull out the OTP or link), built for testing sign-up and verification flows.
- Routes on measured health instead of a hand-kept table. Every call feeds per-provider latency and failure stats, and a weekly CI canary quarantines any backend whose response schema drifts and opens an issue with the diff.
- Every adapter runs one shared contract suite, offline against recorded fixtures and live against the real service. **1,169** offline tests, plus a live test that sends real mail to each provider and reads it back.

OutN | **Python, discord.py, TensorFlow**

- Self-hostable FOSS Pokemon recognition bot for Discord, maintained across **2+ years** and **9 major versions** of breaking upstream API changes without manual migration for existing deployments. Also ships as a desktop app.

EDUCATION

SRM Institute of Science and Technology 2025 - 2030
M.Tech (Integrated) Computer Science, Cybersecurity & Digital Forensics

Delhi Public School 2011 - 2024
Class XII

SKILLS

Languages	Python, Rust, TypeScript, JavaScript, Go, SQL
Backend	FastAPI, SQLAlchemy, PostgreSQL, Hono, Express
Frontend	Next.js, React, Svelte, Astro, TailwindCSS
Infrastructure	Docker, GitHub Actions, Vercel, Supabase, Redis, Git